

TEXTILES MARKET SITUATION REPORT

2024



About WRAP

WRAP is a climate action NGO working around the globe to tackle the causes of the climate crisis and give the planet a sustainable future.

Our core purpose is to help you tackle climate change and protect our planet by changing the way things are produced, consumed, and disposed of.

Introduction

WRAP's Textiles Market Situation Report (TMSR) debuted in 2016, followed by a second edition in 2019, delivered in-depth insights on economic trends impacting the capture, re-use, and recycling of key materials. Since then, the textiles industry has faced several challenges, including the UK's exit from the EU, the Covid-19 pandemic, and a cost-of-living crisis.

These macroeconomic developments, coupled with growing concerns about climate change and the environmental impact of fashion, have questioned established market knowledge and necessitated business practice shifts. Hence, it is timely to assess the current UK textiles market, considering these factors and their implications for sustainability and used textiles.

This report provides a snapshot of the present state of the UK textiles market, enabling the identification of future opportunities and risks, including updated material and economic flows with a focus on sustainability.



Note

In this report, 'textiles' includes clothing, items such as shoes, bags and belts, as well as household-type textiles such as bed and table linen, and leisure textiles such as sleeping bags. Anything outside of this scope will be highlighted in the text or footnotes.

Key Themes and Findings

UK Consumption

The covid-19 pandemic had a large impact on textiles consumption, with rates falling 330kt between 2019 and 2020, however this downward trend was temporary. **Consumption figures are back to near pre-covid levels**, with the UK consuming **1,420 kilotonnes of virgin products** in 2022.

The environmental impact of textiles consumption in the UK remains very high, with carbon and water footprints at **32.9 million tonnes of CO_2e** and **4.98 billion m3 of water**.

UK Manufacturing

The number of textiles and apparel manufacturers within the UK has steadily grown, totalling 9,315. Although investment in Gross Fixed Capital Formation (GFCF) in the textile manufacturing sector has seen reductions, the sales value of manufactured textiles has increased and **the technical textiles sector is expected to have year-on-year growth of 4.36% until 2030**.

Leading up to 2021 **there was a significant decline in the value and volume of textile imports and exports**, largely due to Covid-19 and Brexit. However following 2021, **the volume of imports has recovered by only 12.5%**. Yet, the cost of those imports is greater than pre-pandemic levels, indicating a shift towards higher £/kg ration.

Post-consumer Textiles

Approximately 1,450 kilotonnes of post-consumer textiles were generated in 2022.

Around 727.7 kilotonnes of textiles were discarded to landfill or incineration, with 546.4 kilotonnes disposed of in household residual waste bins, equating to about 4.2% of general waste.

650 kilotonnes of PCTs generated were diverted to reuse and recycling from end-of-life disposal, where **421.6 kilotonnes were exported**.

See Sankey diagram on page 20 for full Post Consumer Textiles flows.

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Glossary/Acronyms

B2C – Business to Consumer.

B&R – Brand and Retailer.

CAGR – Compound Annual Growth Rate. The mean annual growth rate of an investment over a specified period of time, longer than one year.

Chemical polymer recycling -A type of chemical recycling which takes materials back to their polymer level, purifies and reprocesses into pulp or material to be re-spun into new fibres. This includes pulping, solvent-based and hydrothermal processes.

Chemical monomer recycling - A type of chemical recycling, which takes materials back to their monomer and/or oligomer building blocks, purifies and repolymerises for use in place of virgin polymers. This includes Methanolysis, Glycolysis, Hydrolysis and Enzymatic processes.

Commodity code – Commodity codes are internationally recognised reference numbers. A code describes a specific product when importing or exporting goods.[1]

CPI – Consumer Price Inflation is the rate at which the prices of goods and services brought by households rise and fall; it is estimated using consumer price indices.

EoL – End of Life. Sent to waste management facilities for either incineration, landfill or other end of life methods.

GFCF – Gross Fixed Capital Formation. The acquisition of produced assets (including purchases of second-hand assets), including the production of such assets by producers for their own use, minus disposals. Also called “investment”.

HWRC – Household Waste Recycling Centre.

HS – Trade Harmonized System. A standardised numerical system for the classification of traded products.

Mechanical recycling – A type of recycling, where materials are mechanically processed to recover and use the existing fibres in new textiles. This includes processes such as cutting, shredding and flocking.

OECD – Organisation for Economic Co-operation and Development. An intergovernmental organisation founded to promote world trade and economic progress.

P2P – Peer to Peer.

Post-consumer textiles – Textiles that have been purchased, used and then discarded for reuse or disposal. It refers to textiles generated by households or by commercial, industrial, and institutional facilities in their role as end-users. This excludes returned items

Pre-consumer textiles - Textile stock that has been manufactured for and transferred to a retailer but is yet to be used by a consumer. This can include unsold products, returns, and damaged or faulty goods.

R&R – Reuse and Recycler.

Raw Wool – Grease wool (wool as it is shorn from the sheep, before any processing.) in a natural state before scouring.

Scouring – The act of separating dirt, grease and foreign matter from grease wool.

Technical Textiles – Textile products and materials used for functional properties or technical performance over aesthetic or decorative characteristics.

TMSR – Textiles Market Situation Report.

TRA – Textile Recycling Association.

WDF – Waste Data Flow.

WRAP – Waste Action Resource Programme .

Wool Clip – Total amount of wool produced and shorn from a particular flock or flocks in a particular region or country, in one year.

1 https://www.gov.uk/guidance/finding-commodity-codes-for-imports-or-exports#:~:text=Commodity%20codes%20are%20internationally%20recognised,in%20the%20Trade%20Tariff%20tool.



TEXTILE CONSUMPTION

Over the last five years the textiles industry has encountered disruptions in consumption patterns and market dynamics, particularly shaped by the Covid-19 pandemic and the cost-of-living crisis.

UK Consumption

Consumption Trends

Consumption of new textiles fell by 330 kilotonnes from 2019 to 2020.

This considerable downturn can be primarily attributed to the widespread impact of Covid-19. While some recovery was observed, consumption levels by 2021 remained 15% below those recorded in 2019. This indicates a gradual alleviation of the pandemic's influence; however, the persisting cost-of-living crisis in 2022 has created a slow recovery of consumption with figures reaching 1,420 kilotonnes[2,3].

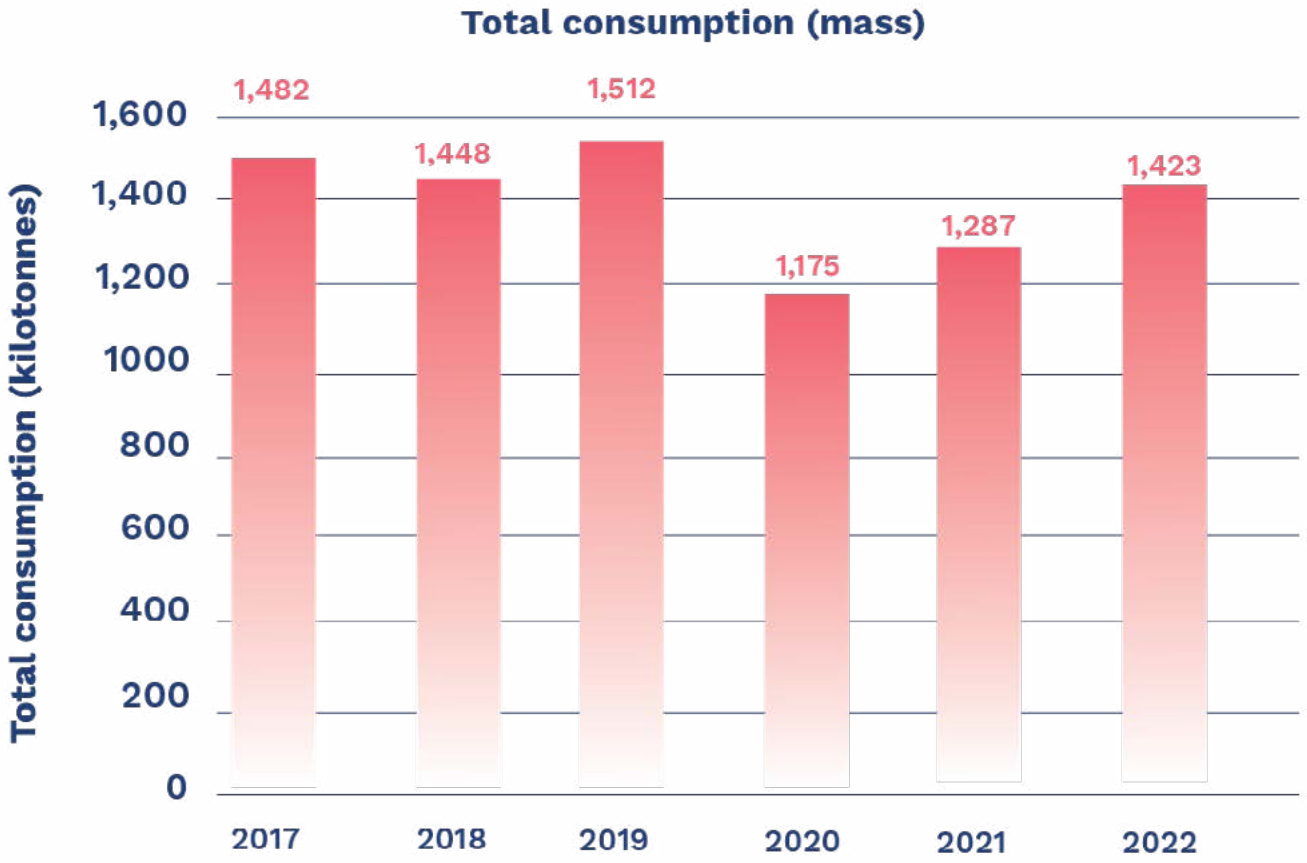


Figure 1: Total UK manufacturer sales of textiles 2011-2021 (UK) (in 1,000 GBP).

2 https://www.uktradeinfo.com/

3 https://www.bankofengland.co.uk/bank-overground/2022/how-has-the-increase-in-the-cost-of-living-affected-uk-households

Household Expenditure

Regarding household expenditure, spending on new clothing (excluding second-hand purchases) underwent fluctuations during the examined period[4]. In 2019, expenditure amounted to £59.1 billion, which decreased to £48.3 billion in 2020, reflecting the impact of Covid-19 restrictions. By 2022, expenditure on new clothing experienced a slight rebound, reaching £56.1 billion[5].

Notably, the Covid-19-induced upheavals led to a significant reduction in the average weekly household expenditure on clothing and footwear[6,7] but expenditure on household textiles witnessed an increase between 2020 and 2021[8]. This is potentially attributable to the Covid-19 restrictions, with consumer shopping habits changing during lockdown[9].

Avg. Weekly Household Expenditure	2019 2020	2020 2021	2021 2022
Clothing and Footwear[10]	£22.80	£14.50[11]	£17.60
Household Textiles[12]	£1.90	£2.20	£2.00

Table 1: Average Weekly Household Expenditure on Textiles in the UK 2019-2022. Data collected from ONS and Statista (2023).

Inflation and Cost

Clothing prices steadily increased by 1.4% from 2015 to 2021[13]. However, in the 12 months leading up to August 2022, the inflation rate soared to 7.6%, significantly hampering the recovery of clothing consumption.

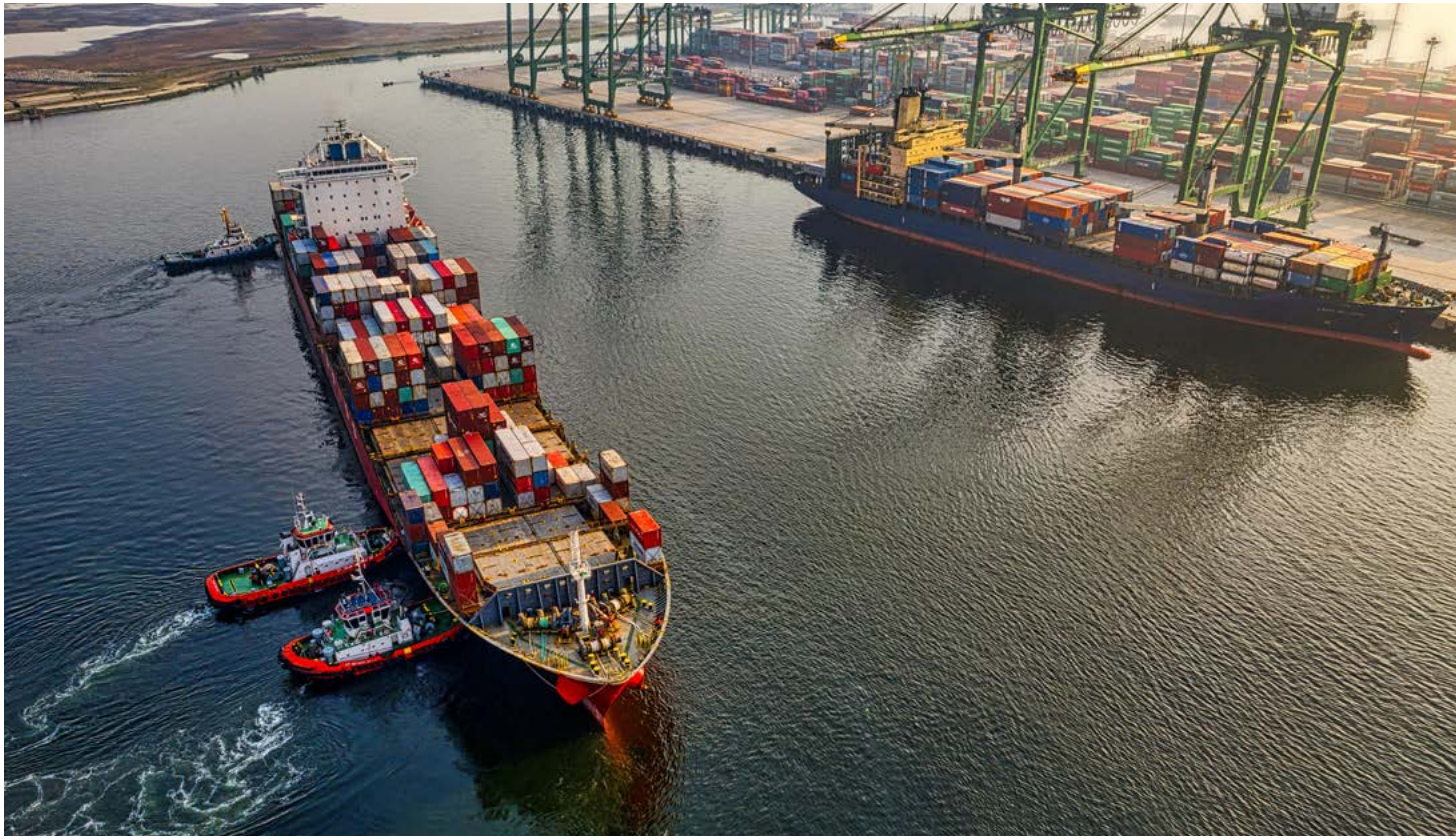
The average cost per clothing item followed a different trajectory, registering nearly a 2% decrease since 2015 and finishing at £16.37 in 2022[14]. Notably, this figure is significantly below the overall Consumer Price Index (CPI) rate during the same period which actually rose by 9% between 2021-2022.

4 Clothing costs consist of clothing materials, garments, other articles of clothing and clothing accessories, and the cleaning and hire of clothing.
5 ONS, Consumer trends: chained volume measure, seasonally adjusted, Consumer trends: chained volume measure, seasonally adjusted - Office for National Statistics
6 Deflated to 2019 figures
7 Family spending workbook 1: detailed expenditure and trends - Office for National Statistics (ons.gov.uk)
8 https://www.statista.com/statistics/913292/home-textiles-weekly-uk-household-expenditure-by-type/
9 https://wrap.org.uk/resources/report/citizen-insights-clothing-behaviours-during-covid-19-restrictions-part-2#download-file
10 Family spending workbook 1: detailed expenditure and trends - Office for National Statistics (ons.gov.uk)
11 Deflated to 2019 figures
12 https://www.statista.com/statistics/913292/home-textiles-weekly-uk-household-expenditure-by-type/
13 https://www.ons.gov.uk/economy/inflationandpriceindices/datasets/consumerpriceinflation
14 Statista. (September 2, 2022). Average price per unit in the apparel market in the United Kingdom (UK) from 2013 to 2026 (in GBP) [Graph]. In Statista. Retrieved December 15, 2022, from https://www.statista.com/forecasts/1156118/price-per-unit-in-apparel-market-in-the-uk

Imports and Exports

Between 2017 and 2021, the UK textiles industry experienced a significant decline in imports and exports. Import volumes dropped by nearly 590 kilotonnes or £5 billion, with a notable downturn in 2020 due to Covid-19 and Brexit. Although there was a slight increase in import mass in 2021, the overall value continued to decline (£3.2 billion), indicating a shift towards lower-priced textiles.

However, between 2021 and 2022 as the volume only increased by value 74 kilotonnes, the value increased by £6.3 billion, showing a shift in towards items with a higher £/kg ratio. Textiles exports decreased by over 480 kilotonnes or just over £3 billion, primarily driven by reduced exports to the European Union (EU). The most significant decline in export value occurred from 2020 to 2021 (£2.7 billion), with a sharp drop in EU exports. The decline in both imports and exports can be attributed to the combined impact of Covid-19 and Brexit.



Top Ten Import Countries

The map and accompanying chart below show the top ten countries that the UK imports consumer textiles from.

Country	Import Mass (Tonnes)	Import Value (Millions £)
China	417,600	5,500
Bangladesh	232,700	3,200
Turkey	75,600	1,500
India	85,100	1,400
Pakistan	137,700	1,100
Vietnam	36,600	800
Cambodia	38,300	600
Sri Lanka	29,100	500
Myanmar	27,500	400
Hong Kong	10,300	300

Table 2: Top Importers (by value) and associated mass. Source: UK Trade Info (2022).

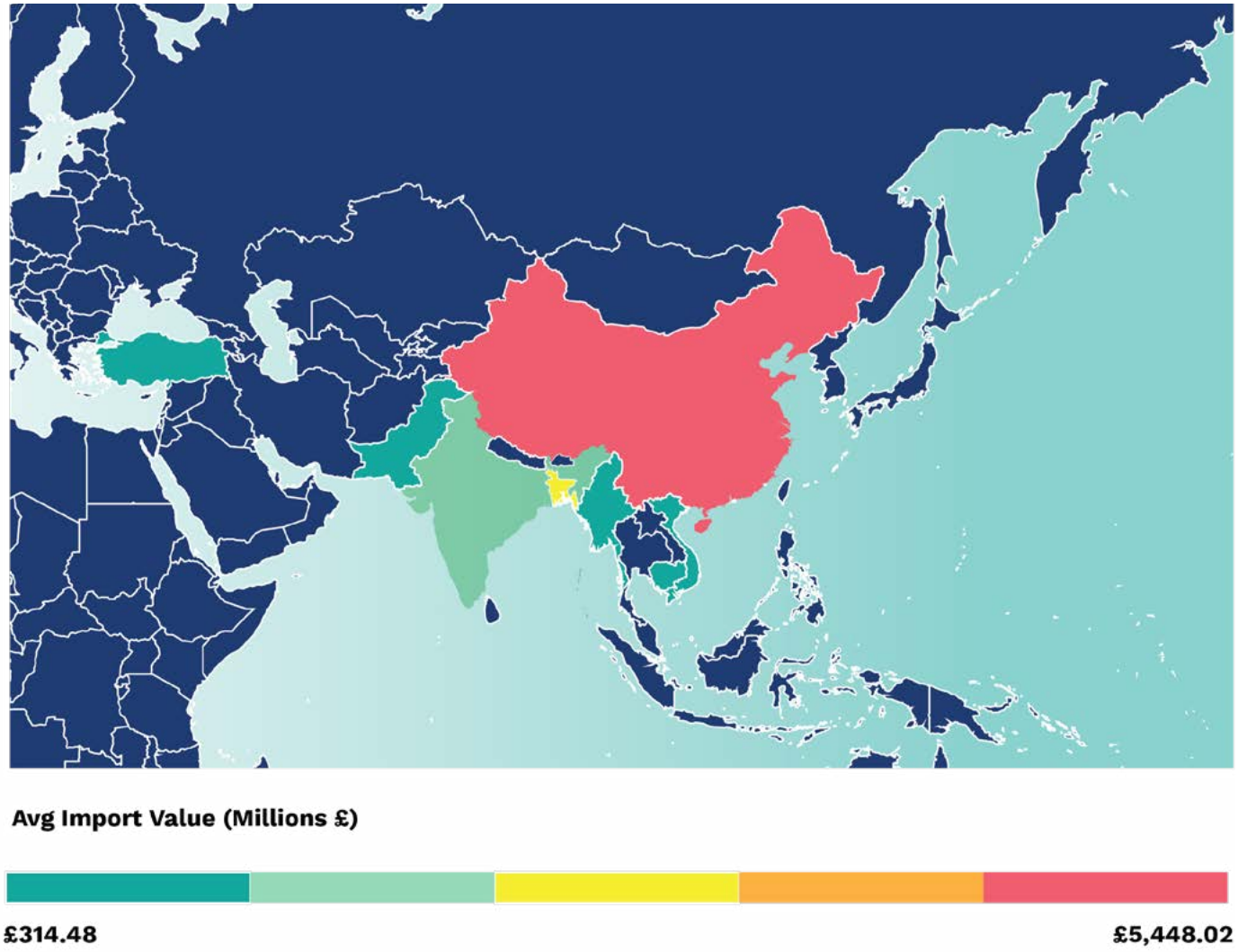


Figure 2: Top Ten Import Countries by Value. Source: UK Trade Info (2022).

Top Ten Export Countries

The map and accompanying chart below show the top ten countries that the UK exports consumer textiles to

Country	Export Mass (Tonnes)	Export Value (Millions £)
United States	7700	560
South Korea	700	400
Hong Kong	950	200
Australia	4500	150
UAE	3200	100
Macao	76	100
Japan	800	90
Switzerland	560	85
Saudi Arabia	1200	65
China	1200	50

Table 3: Top Exports (by value) of Textiles from the UK. Source: UK Trade Info (2022).

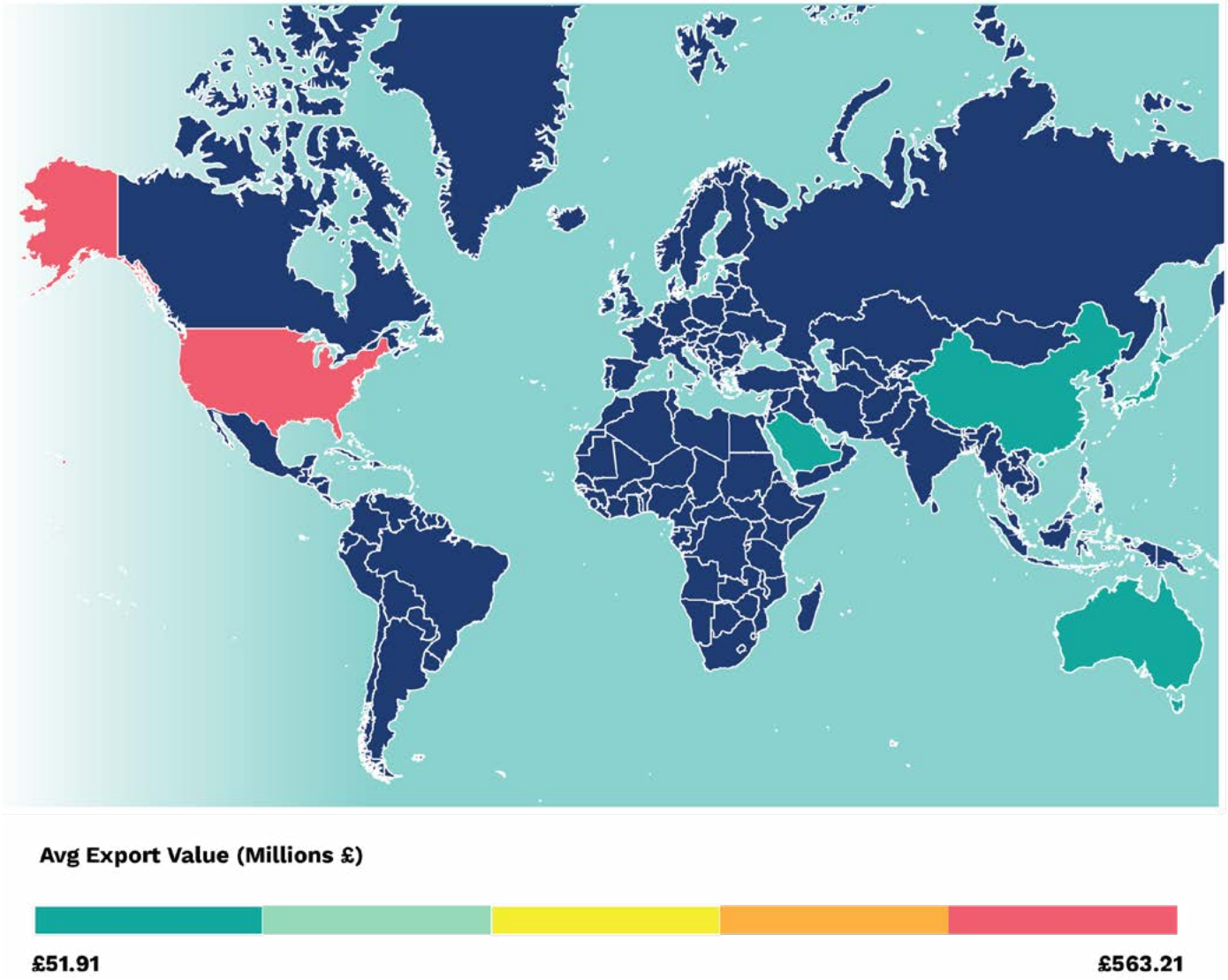


Figure 3: Map of the Top Ten Export Countries by Value. Source: UK Trade Info (2022).

The carbon and water footprints of textiles consumed in the UK decreased from 2019 to 2021, but raised again in 2022

Environmental Impact

The carbon and water footprints of textiles can be mitigated through various measures including using less impactful or recycled fibres and extending the lifespan of clothing. The carbon and water footprints of textiles consumed in the UK was estimated using the Textiles 2030 Footprint Tool[15]. The tool measures the environmental footprint for the full life cycle impact of textiles. This is based on Textiles 2030 signatories that cover 62% of the market[16]. **The carbon and water footprints of textiles consumed in the UK decreased from 2019 to 2021, but raised again in 2022 (see table 4 and 5)**. This can be attributed to a decrease in the mass of textiles consumed during that period due to Covid-19. However, **the impact per tonne of fibre used in the UK changed by less than 1% for both carbon and water footprints.** To further reduce these impacts, decision-makers in the UK can consider increasing the use of 'improved' or 'preferred' fibres as referenced in the WRAP Circular Design Tool Kit[17] or reducing the total quantity of textiles produced and imported.

15 wrap.org.uk/resources/report/textiles-2030-footprint-tool
16 wrap.org.uk/resources/report/textiles-2030-annual-progress-report-202122#download-file
17 https://wrap.org.uk/resources/guide/circular-design-toolkit

	2019	2021	% decrease
Carbon Footprint	34.5 million tonnes CO_2e	29.4 million tonnes CO_2e	14.8%
Water Footprint	5.64 billion m^3	c. 4.67 billion m^3	17.2%

Table 4: Carbon and water footprints of textiles in the UK 2019 versus 2021.

	2021	2022	% increase
Carbon Footprint	29.4 million tonnes of CO_2e	32.9 million tonnes of CO_2e	11.9%
Water Footprint	c 4.67 billion m3	c. 5 billion m^3	7%

Table 5: Carbon and water footprints of textiles in the UK 2021 versus 2022.

Fibres	2019	2021	2022
Cotton	349,700	282,400	313,064
Wool	15,800	11,800	14,230
Silk	20	1,930	1,138
Flax/linen	88,900	75,700	73,997
Viscose	96,800	82,300	101,034
Polyester	786,800	693,200	768429
Acrylic	39,900	19,300	19,922
Polyamide	75,400	64,200	71,151
Polyurethane/ Polypropylene/ Elastane	39,900	34,800	54,075
Other	14,000	18,100	5,977

Table 6: UK textiles apparent consumption by fibre type for 2019-2022 (tonnes)



UK MANUFACTURING

Outlining the UK's textile and apparel manufacturing landscape, showing growth trends, challenges, and the promising expansion of technical textiles

UK Manufacturing

There are a combined total of 9,315 textiles and apparel manufactures within the UK[18]

Manufacturing businesses have been steadily increasing since 2015, with 870 new enterprises opening within the eight-year period.

Manufacture of textiles

Sales value of textile[19] manufacturing in the UK increased from £3.94 billion in 2010 to £4.38 billion in 2021, slightly down from the peak of £4.53 billion in 2019[20].

The increase was mainly driven by:

- An increase in the value of 'Manufacture of made-up textiles articles, except apparel' e.g. bed linen (£411 million)
- An increase in the value of 'Manufacture of non-wovens and articles made from non-wovens except apparel' e.g. non-woven fabrics (£128 million).

During the same period, the value of 'Preparation and spinning of textile fibres,' 'Finishing of textiles,' and 'Manufacture of knitted and crocheted fabrics' decreased by £112 million, £195 million, and £56 million, respectively. In 2019 to 2020, there was a significant decrease of £550 million in the value of textile manufacturing, likely due to complications and reduced demand caused by the Covid-19 lockdowns.

Manufacturing locations

The UK has a total of 4,435 textiles manufacture enterprises as of 2022[21]. With England holding the most (85.7%), followed by Scotland (7.0%), Wales (3.9%) and Northern Ireland (3.4%). Within England, most of the textiles manufacturing enterprises are based within the North West of England (15.4%), yet East Midlands (14.2%), London (13.9%), the South East (13.2%) and Yorkshire and The Humber (12.5%) closely follow.



Figure 4: Revenue of the textiles market in the UK (in billion GBP) – Sources: HMRC and ONS (2022)

18 www.ons.gov.uk/businessindustryandtrade/business/activitysizeandlocation/datasets/ukbusinessactivitysizeandlocation

19 The Textiles are covered by Division 13 – Manufacture of Textiles. This division includes preparation and spinning of textile fibres as well as textile weaving, finishing of textiles and wearing apparel, manufacture of made-up textile articles, except apparel (e.g. household linen, blankets, rugs, cordage etc.).

20 https://www.ons.gov.uk/businessindustryandtrade/manufacturingandproductionindustry/datasets/ukmanufacturerssalesbyproductprodcom

21 https://www.ons.gov.uk/businessindustryandtrade/business/activitysizeandlocation/datasets/ukbusinessactivitysizeandlocation

Manufacture of clothing

The sales value of wearing apparel manufacturing in the UK reached its lowest level in 2021 (£1.27 billion) since 2010 (£1.53 billion)[22]. There was a significant reduction in sales from 2019 (£1.61 billion) to 2020 (£1.38 billion). The sectors that saw a decline were;

- 'Other outerwear' (-£93 million)[23]
- 'Underwear' (-£183 million)
- Knitted and crocheted hosiery (-£82 million) manufacturing
- Sales of 'Leather clothing' and 'Workwear' also declined £2.1 million and £39.8 million respectively

Yet, in this period 'Other wearing apparel and accessories[24] and 'Other knitted and crocheted apparel' did see an increase of £125 million and £19 million, respectively. Unlike clothing consumption, there was no recovery in manufacturing sales from 2020 to 2021, and they continued to decrease.

The 'manufacture of other outerwear' is the largest subsector in the UK representing 44.3% of UK apparel manufacturing.

Fibre prices

In 2020, most fibres experienced a drop in prices, with lower prices in Q2 2020 compared to Q2 2019 (except for Spandex, which slightly increased). However, prices began recovering in 2021 and 2022, surpassing 2019 levels for several fibres in Q2 of those years.

Cotton prices hit a multiyear low in 2020 but increased in 2021. Stable cotton prices are expected in 2023, due to reductions in consumption caused by the negative global economic outlook and high inflation being somewhat offset by reductions in cotton production[25]. Potential polyester price increases in 2023 will be limited if crude oil prices remain low. The overall negative global economic situation may hinder significant price increases for textiles, while volatility and decreased business confidence could cause price decreases.

Fibre prices have fluctuated since the end of 2017

Cotton, acrylic, viscose, spandex, and flax saw price increases of over 10% between Q3 2017 and Q1 2023 (with flax increasing by over 100%).

Conversely, polyester, nylon, and wool prices decreased by at least 9% from Q3 2017 to Q1 2023

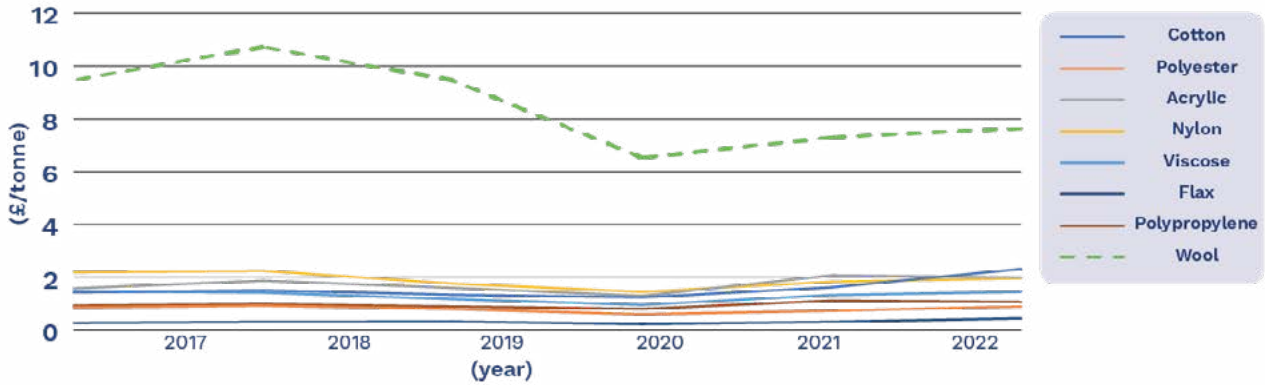


Figure 5: Annual fibre prices 2017 – 2023 (£/kg) - Source - TexPro: Textiles & Apparel, historic prices (2023)

22 https://www.ons.gov.uk/businessindustryandtrade/manufacturingandproductionindustry/datasets/ukmanufacturerssalesbyproductprodcom

23 'Other outerwear' - e.g. Coats, suits, skirts etc

24 'Other wearing apparel and accessories' – Other wearing apparel e.g. Babies' garments, tracksuits, swimwear etc. Accessories e.g. gloves, belts, shawls, hairnets etc

25 Global cotton consumption to fall below output after 2 years: ICAC - Fibre2Fashion

UK wool manufacture

Production of raw wool in the UK has been stable during the last decade with a slight increase in production between 2010 and 2021 going form 67 kilotonnes to 70.4 kilotonnes[26]. The main uses for raw wool in the UK are bedding and other homeware, clothing, yarns for knitting and crochet, carpets and insulation[27].

Prices of wool clip, on the other hand, have had heavy fluctuations, with a very pronounced decline from 2011 to 2021[28] due to a reduction in demand for woollen fabrics in the last two decades as a result of manmade fibres and other natural fibres that are more economically feasible such as acrylic, polyester and cotton[29].

The volume of wool imported into the UK in 2021, which was in the form of non-manufactured textile wool not carded or combed, was slightly over 36 million kilograms[30], with a value of $57.7 million[31]. By comparison, the amount of non-manufactured textile wool exported from the UK in 2021 was of 20.5 million kilograms[32] with a value of almost $46 million.[33]

Technical textiles

Technical textiles are textile products and materials used for functional properties or technical performance over aesthetic or decorative characteristics.

The UK technical textiles sector is experiencing strong growth. In 2022, it generated £9.68 billion and is projected to have a compound annual growth rate (CAGR) of 4.36% until 2030, reaching a forecasted value of £13.39 billion. In 2022, 2,394 kilotonnes of technical textiles were manufactured, dominated by woven material and synthetic polymer fibres[34].

The number of businesses in this sector accounts for 5.64% of UK textiles manufacturing, with a notable growth of 74% between 2017 and 2022[35]. The growth is primarily driven by domestic automobile production and the demand for specialized components. The largest application of technical textiles is in the transport sector, followed by industrial products and components. Medical and hygiene has the highest CAGR at 5.64%.



26 UK wool annual production 2021 | Statista
27 What do we use wool for? | British Wool
28 Wool clip output price trend UK 1994-2021 | Statista
29 science behind the wool industry. The importance and value of wool production from sheep | Animal Frontiers | Oxford Academic
30 UK wool imports 2021 | Statista
31 UK wool import value 2021 | Statista
32 UK wool exports 2021 | Statista
33 UK wool export value 2021 | Statista
34 Verified Market Research, 2022, UK Technical Textiles Market: Market Size, Status and Forecast to 2030
35 https://www.ons.gov.uk/businessindustryandtrade/manufacturingandproductionindustry/datasets/kmanufacturerssalesbyproductprodcom



POST-CONSUMER TEXTILES MARKET

The following section will cover multiple areas of the post-consumer textiles market such as textiles collected for diversion in the UK; textiles sold/traded in the UK and textiles sent for end of life[36].

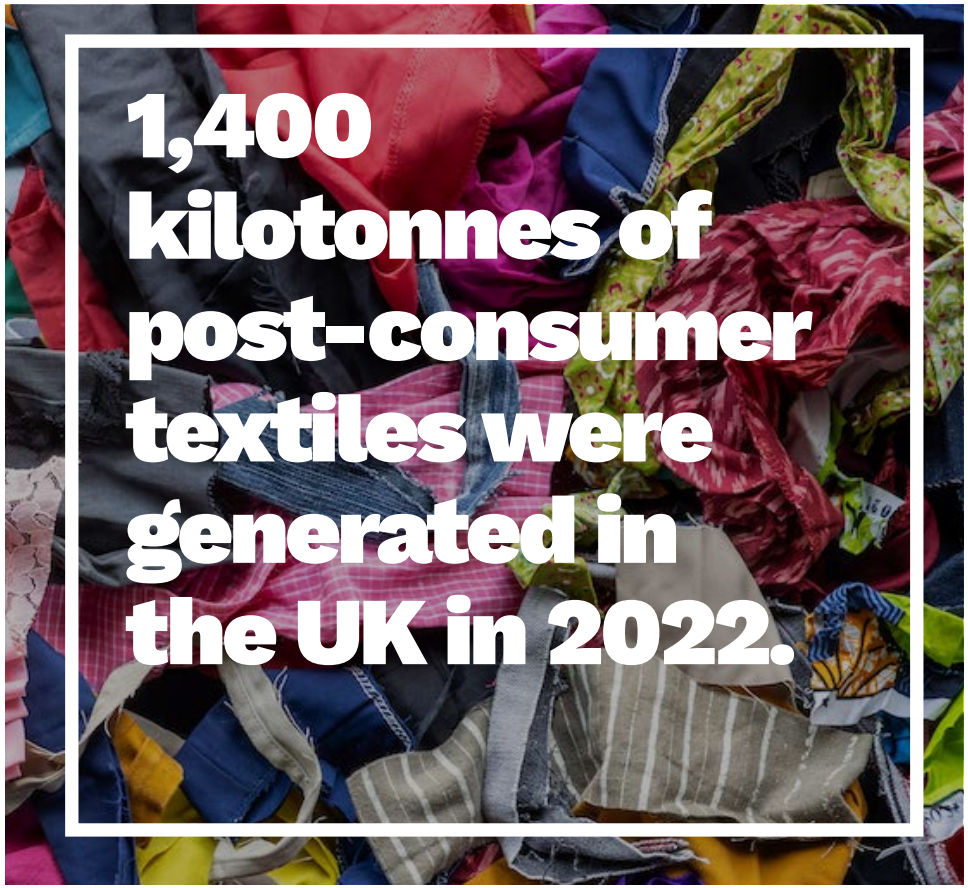
36 The analysis covers clothing, home textiles, leisure textiles, and accessories (shoes, bags, belts). Other categories like mattresses, carpets, workwear, pre-consumer clothing, post-industrial waste, etc., are not included.

Post-consumer textiles generated

Around 1,450 kilotonnes of post-consumer textiles were generated in the UK in 2022. See Sankey diagram on page 20 for the post-consumer materials flows and Table 7 for breakdown of key volumes by stage.

Stage/Activity/Actor	Estimated quantity 2021 (Tonnes)[37]
PCTs diverted from end of life	650,000 (Up 5% from 2019)
PCTs sold to the public	276,000
PCTs sent for end of life	759,000

Table 7: Quantification (tonnes) of post-consumer textiles at different stages and routes in the UK market in 2022.



37 Note, the total does not add to 1,450 kilotonnes. This is due to double counting in the onward destinations and material flows following PCT generation e.g., tonnages diverted by charities may be sold to public or could be sent for end of life.

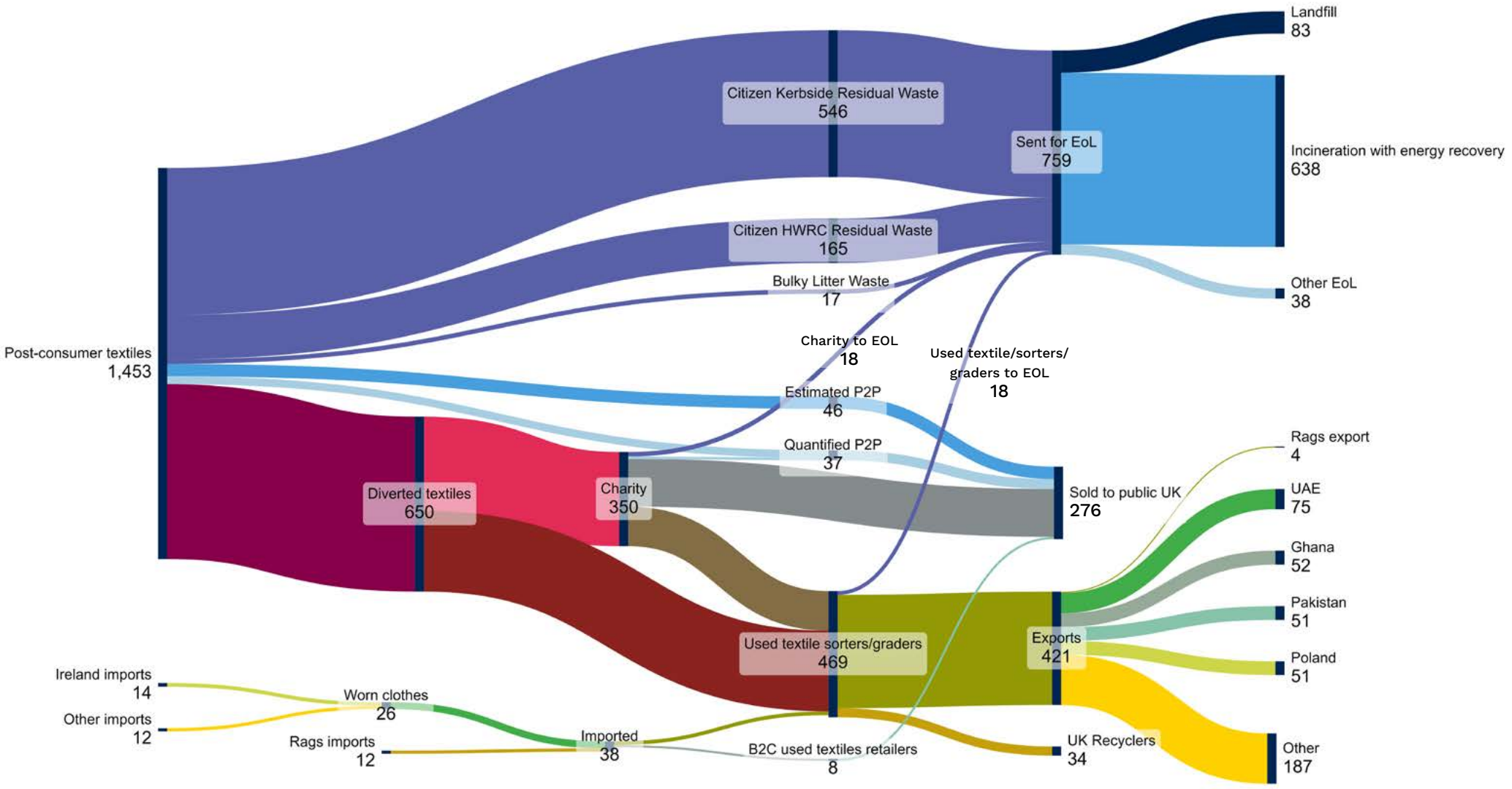


Figure 5: Sankey diagram showing the UK market for used textiles by kilotonnes, estimated value (£ million).

Post-consumer textiles diverted from end of life

Post-consumer textiles for diversion	Volume (Tonnes)
Local authority managed collections (kerbside/HWRCs/bring banks)	240,900
Charities	350,000
Textile collectors/sorters & graders	469,000

Table 8: Volume (tonnes) collected for diversion by key reuse and recycling actors.

Textile collectors/sorters & graders

For this report, textile collectors, sorters and graders are combined together as one business type, as many actors participate across the reuse and recycling value chain, both collecting and sorting.

Textiles collectors, sorters & graders collect and handle the greatest volumes of used textiles by business type around 469 kilotonnes of used textiles. See Figure 5 for pie chart with list of sources and percentage breakdown. The majority (45%) of used textiles is sourced from privately operated commercial bring banks. Following this, an estimated 32% are collected from charities – this includes shops and charity managed bring banks, which are then sorted and graded in preparation for the next destination.

Average price per tonne of used textiles collected in 2022 runs at £329[38]. The monetary value of used textiles is highest when the source of the collected textiles was from Local Authority kerbside collections. Bring banks and HWRCs were found to be of the lowest value[39].

Brand and retailer partnerships and take-back schemes

Brand and Retailer (B&R) take-back schemes are a growing source of post-consumer textiles, as companies attempt to build on more circular and sustainable ways of disposing used garments. Take-back schemes are usually managed by third parties such as charities, collector/sorters or online resellers. Data reports to Textiles 2030[40] has shown there has been a 60% increase in the volume of used textiles collected from signatory take-back schemes between 2019 and 2021. In 2021, signatories collected over 45 kilotonnes of used textiles from online and in-store take-back schemes, with this number expected to continue to rise with the uptake of schemes by B&R.

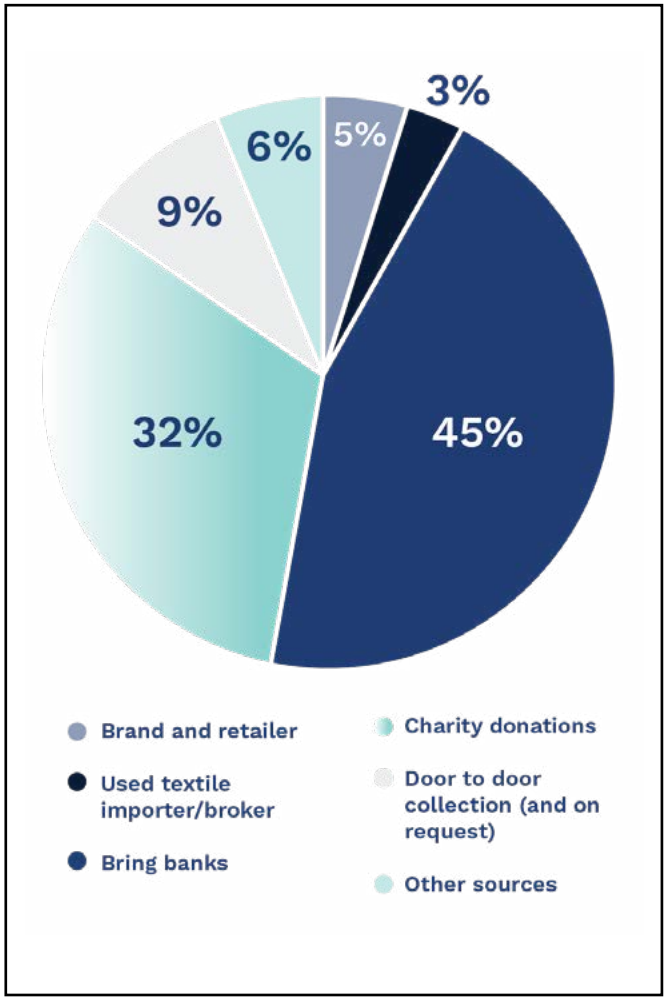


Figure 6: Pie chart displaying used textile collectors/sorters & graders sources and percentage breakdown.

38 WRAP, Unpublished, Textiles Market and Waste Hotspot survey findings
39 WRAP, Unpublished, Textiles Market and Waste Hotspot survey findings
40 WRAP, 2022, Baseline Report

Post-consumer textiles sold to the public

Approximately, 276 kilotonnes of post-consumer textiles were sold to the public via various streams in 2022, including by charity retailers, business to consumer used textiles retailers and peer to peer (P2P) reuse organisations – see Figure 5 for breakdown. Research suggests sales of post-consumer textiles have increased in past years, with notable changes also seen in the method of sales (e.g., online/P2P platforms/in store), however there is variation by business type.

From research and reporting by consulting[41] and market research firms[42], the total value of the UK second-hand clothing market ranges between low estimates of £3,200 million and high estimates of £6,500 million, this averages out at £4,845 million. Note, there is a lack of reliable data in this area, this figure should act as an estimation only.



Understanding how the differences between sources of used textiles impact used textiles value would support the business case for textile reuse, and the implementation of strategies to promote collection from specific sources offering the highest value option.

41 Oakdene Hollins calculation (low estimate)

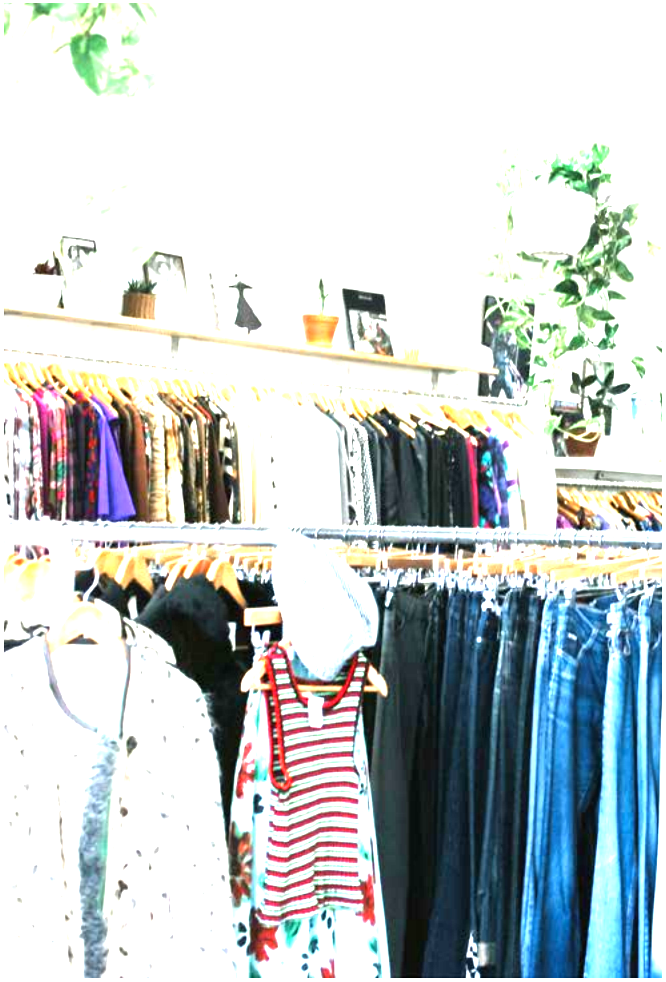
42 Future Market Insights, 2022, Secondhand Apparel Market Outlook (2022-2032),

Charity retailers

According to the Charity Retail Association approximate 53% of textiles (c. 185.5 kilotonnes) collected by charities are sold in store or online[43].

Apparel makes up 65% of sales for charity retailers, worth an estimated £552.8 million in 2022. Industry experts indicated in interviews with WRAP that charity retail revenue is at an all-time high, largely due to improved commercial and retailing practices that are increasing the price of an average item (see Table 8).

Average prices of womenswear and menswear stand at £3.95 and £4.25 per item, respectively. Charities are also selling textiles through e-commerce, with around 4% of total retail sales made online, predominantly on eBay (65.6%), followed by charities' own websites (33.7%) and other reuse sites (0.7%)[44]. The volume of used clothes sold in charity stores has decreased, with more items being sent to used textile collectors due to lower quality. Lower quality items are less desired by customers and bring in a smaller margin for charities and therefore are more challenging to sell.



Item	Average Price (£)
Womenswear (average)	£3.95
Womenswear blouses, tops, knitwear	£3.89
Womenswear skirt and trousers	£3.82
Menswear (average)	£4.25
Menswear shirts, t-shirt, knitwear	£3.90
Menswear trousers	£4

Table 9: Average price of clothing sold by charity retailers by product type[45].

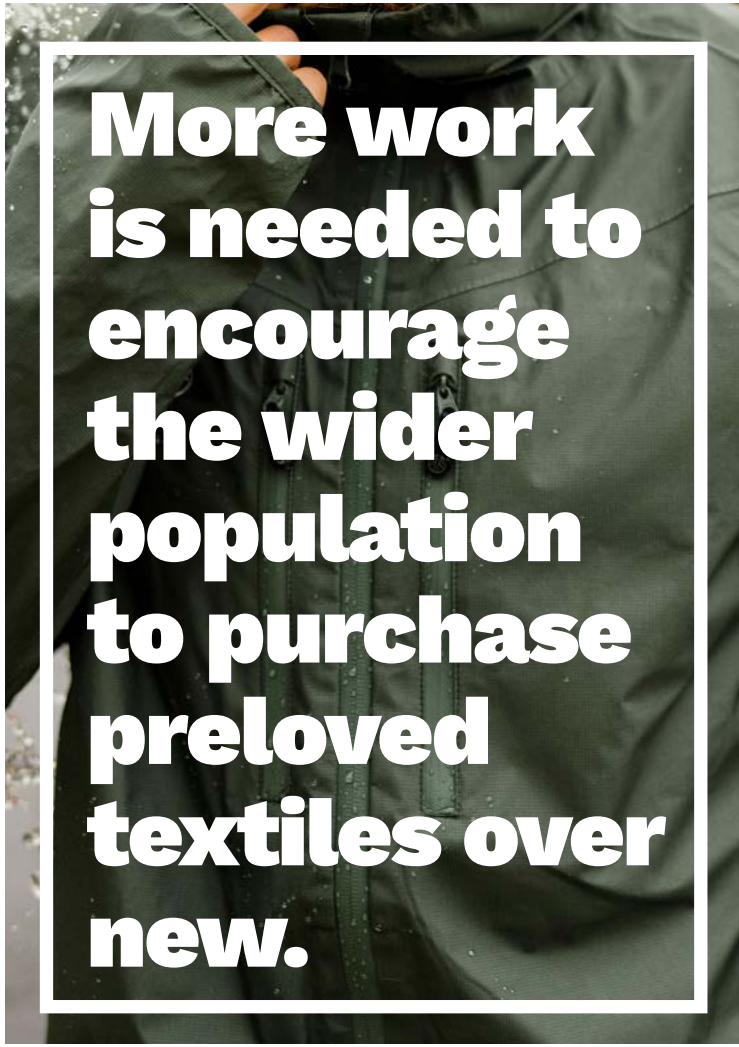
43 WRAP, Unpublished, Textiles Market and Waste Hotspot survey findings
44 WRAP, Unpublished, Textiles Market and Waste Hotspot survey findings
45 https://www.charityretail.org.uk/key-statistics/

Peer to peer reuse organisations

P2P reuse organizations, including digital platforms and consignment stores, connect sellers and buyers directly without owning the goods. These platforms also attract charities and B&R organisations for selling pre- and post-consumer textiles. The use of online P2P platforms has increased, especially during the pandemic, with a 9.7% revenue growth in e-commerce between 2021 and 2022[46].

The volume of textiles traded through P2P platforms is estimated at 82.5 kilotonnes, but accurate quantification is challenging due to limited transaction records[47].

The price per kilogram of used textiles on P2P platforms varies based on retail channels, product type, and brand[48]. It is unclear if items sold through P2P platforms are in addition to, or diverted from, traditional reuse channels. However, if they are in addition to traditional channels this suggests great potential as currently on P2P platforms more items are listed for sale than actually sold[49]. Meaning they've been able to tap into the 26% of unworn clothing currently dormant in the publics wardrobes[50].



46 Oberlo, 2022

47 For the exercise, 3 platforms considered were eBay, Depop, and Vinted, due to their relative size to the market and the publicly available data. Of these platforms, the volume of textiles trade are approximately 36,500 tonnes. Unquantified P2P volumes are estimated at 46,000 tonnes.

48 eBay is considered to be the P2P market leader, where the price per kilogram of textiles is £39.70. This figure includes apparel, footwear and accessories, but may not include home and leisure textiles.

49 Morning Consult, 2022, What Retailers Need to Know About Resale's Buyers and Sellers. The research did not focus solely on textiles or P2P platforms.

50 WRAP,2022. Citizen Insights: Clothing Longevity and Circular Business Models Receptivity in the UK.

Recovered textile prices

Covid-19 saw recovered textile prices impacted, with prices for textiles banks bottoming out around mid-2020 at £15 per tonne, due to disruption and declining demand due to lockdowns. This period also saw nationwide closure of charity shops, making price data inaccessible. The value of recovered textiles stabilized in mid-2021 but at lower levels, The UK's economic slowdown in late 2022 has kept prices low due to rising costs and reduced discretionary spending.

As of January 2023, textile bank prices are £172.5/tonne, and charity shop prices are £255/tonne, encouraging the collection of used textiles for reuse and recycling.



Figure7: UK Recovered Textiles prices (£ per tonne). – Source: WRAP Materials Pricing Report (2023).

Business to consumer used textiles retailers

Research suggests the volume of textiles sold by business to consumer used textiles retailers in the UK in 2022 was 7,800 tonnes[51]. To help compare, this is a tiny fraction of both the post-consumer textiles generated (0.6%) and post-consumer textiles sold (3%). WRAP estimates the average price per kilogram

for used textiles from this business type is £56.70, significantly the highest price for used textiles within the reuse market. The value of the business to consumer used textiles retailer market is estimated at £441.2 million in 2022[52]. Note, there is a large range of uncertainty, as prices per kilogram and markups on these vary widely depending on the method to commercialise the used textiles." (see page 28 on the word report)

Rental organisations

The UK saw significant growth in the clothes rental market between 2021 and 2022, with **an estimated volume of 135.2 tonnes rented annually[53].** This corresponds to approximately 213,100 units and 2.8 million rentals per year. New rental business models, including peer-to-peer renting, have emerged. However, the current estimate suggests that less than 100,000 items are rented through peer-to-peer platforms.

The rental market relies on two key technologies: logistics & back-end technology and front-end technology. Logistics & back-end technology facilitates the physical flow of rentals, requiring specialised infrastructure and capacity. Front-end technology enables businesses to offer rental services through websites or rental marketplaces. As the rental market expands, more players are expected to enter the market on the fulfilment side, leveraging the ease of front-end technology implementation.



51 WRAP, Unpublished, Textiles Market and Waste Hotspot survey findings. Low estimation 5,190 tonnes, high estimation 10,381 tonnes. Level of confidence low.
52 WRAP, Unpublished, Textiles Market and Waste Hotspot survey findings. Low estimation £51.9 million, high estimation £830.5 million. Level of confidence low.
53 WRAP, Unpublished, Textiles Market and Waste Hotspot survey findings. Estimations based on figures from a market leader

Imported post-consumer textiles

- 38 kilotonnes post-consumer textiles imported in 2022, up from 15.9 kilotonnes from 2021.
- The increase in clothing sales attributed to B2C used textiles retailers, vintage wholesalers, and international shipping from P2P platforms.
- Over half of the volume of used textiles imported into the UK comes from Ireland due to having the lowest value per kilogram (£0.62).
- Poland and the Netherlands are known for supplying vintage clothing to the UK.
- An unknown volume of used textiles are collected in the UK, exported and re-imported after detailed sorting and grading in other countries. Reasons for re-importation of used clothing to the UK include greater clothing volumes than UK sorting capacity, cheaper labour and established infrastructure abroad.

There is a clear need and business case to increasing sorting capacity in the UK.

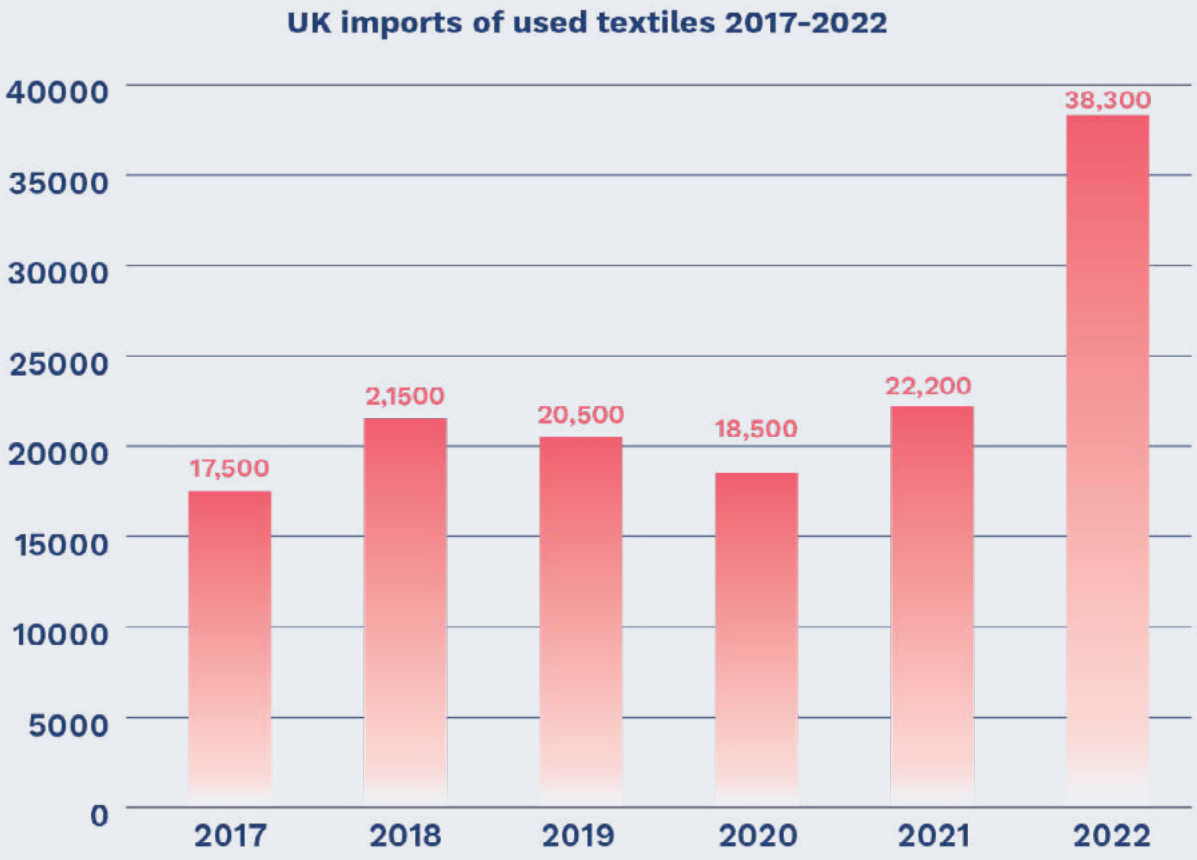


Figure 8: UK Textiles imports into the UK 2017-2022 by mass (Kilotonnes). Source: HMRC (2023).

Exported post-consumer textiles

The UK is one of the world's biggest exporters of used clothing, behind the US and China. The top export destinations in 2022 were the UAE, Ghana, and Pakistan.

- 421.6 kilotonnes of post-consumer textiles were exported from the UK in 2022.
- Exports reached their highest level in 2022 after a dip in 2020 due to Brexit uncertainty and the Covid-19 pandemic.
- 99% of the exports were categorized as worn clothing, and 1% as rags.
- Sorting geographies for used textiles traditionally based within the E.U. and U.S. are shifting to other countries in South America, Pakistan, India, Bangladesh, and the United Arab Emirates.
- The UAE has a fast growing textile sorting and grading industry. Free-trade zones allow unsorted used textiles to be exported from the UK to be sorted in-country.
- The UAE is an example of wider offshoring of sorting and grading operations in the used textiles industry. Sorting is increasingly economically unviable in the UK, due to high volumes of low quality volumes, high labour costs and low economic returns.
- The estimated value of exported textiles in 2022 was £367 million, with an average price of £0.87 per kg.

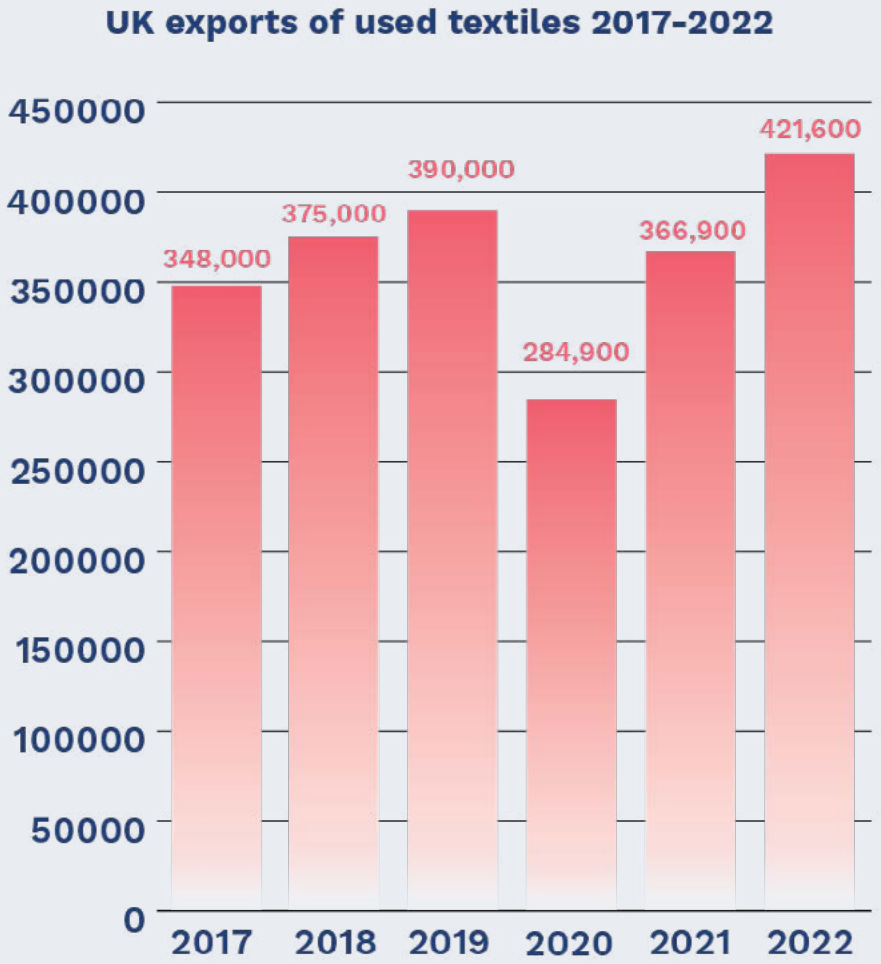


Figure 9: UK Textile Exports in Kilotonnes. Source: HMRC (2023).

End markets

Onward markets refer to markets for post-consumer textiles following collection and sorting

Onward Markets	Volumes (Tonnes)
Exports	421,600
Open/closed-loop recycling	34,000

Table 10: End markets of post-consumer textiles generated included volume (tonnes).



Open/closed-loop recycling

The other onward market for post-consumer textiles collected outside of exports is open/closed-loop recycling.

The current volume of feedstock for textiles recycling is estimated at around 34,000 tonnes.

This mass is mainly direct from used textiles collectors, sorters & graders or charities, with a point-of-sale value at approximately £29.2 million. Per kilogram, costs for the feedstock are similar to exports, estimated to average out at £0.87.

These volumes are then mechanically recycled, where the used textile is typically cut into smaller pieces and utilised in new textile-based products, such as non-wovens and wipers. By grinding and shredding processes, post-consumer textiles can also be returned to its fibrous form for products such as stuffing and insulation. The majority of this recycled product (90%) is sold within the UK. The domestic market is prevalent for manufacturing components and mattress materials, whilst recycled wipers are largely exported.

Fibre to Fibre (F2F) recycling in the UK

F2F mechanical recycling is currently taking place in the UK for wool and cotton. Based off current business developments, by 2023, recycling polyester using a thermo technique is also expected to be operative in the UK.

Volumes that are 'chemically recycled' through chemical polymer or chemical monomer recycling processes in the UK are still very low, estimated at below 1,000 tonnes per year. This accounts to around 2% of textiles recycled into new products in the UK. At the time of writing, no chemical recycling was fully operative, with a small number of chemical recycling businesses currently in the research and development and piloting phases.

Comparatively to Europe and globally, F2F recycling is still in its relative infancy in the UK and requires significant investment to be an economically efficient equivalent to virgin fibres and to operate at scale. Developments are expected in the UK in the next few years, which would likely increase the value per kilo of post-consumer textiles once sorted, to create mass balance suitable for F2F technologies. This represents a substantial market opportunity as well as increased and enhanced UK sorting capacity.

Textiles sent for end of life

Textiles sent for end of life enter residual waste streams such as general rubbish bins, residual waste banks at HWRCs, bulky waste and litter. These textiles end up being disposed of primarily either through incineration or landfill – see Table 7. They are widely measured through the platform Waste Data Flow, as local authorities or their contractors, are responsible for reporting on residual waste collections.

In 2021, there were approximately 727.7 kilotonnes of clothing, shoes, bags and non-clothing textiles from the sources above sent to landfill or incineration in the UK.

This number increases to 759 kilotonnes, when textiles sent to end of life from charities and used textile collectors/sorters & graders are included – see Table 12 for tonnage breakdown. This figure represents over half (54%) of all post-consumer textiles generated in the UK in 2022.

Source	Volume of textiles (tonnes)	% of the total
Kerbside residual waste	546,390	72%
HWRC residual	164,850	22%
Bulky waste	5,140	1%
Litter	11,280	1%
Charity EoL	17,500	2%
Used textiles collectors/sorters/graders	13,840	2%
Total textiles	**759,000**	**100%**

Table 11: Textiles sent to end of life disposal by waste stream and volume (Tonnes).



Textiles found in the residual waste

Household residual waste

Of the textile tonnage found in household residual bins in the UK, the highest proportion is identified as clothing, totalling 47% or an estimated 259.5 kilotonnes. This figure stands at approximately 2% of waste disposed of in the residual stream[54]. See Table for tonnage breakdown by product type.

Please note, comparisons on textiles arisings in residual waste (kerbside and HMRC) against our previous TMSR in 2019 should be taken with caution, due to Covid-19 impacting collection operations during research years.

Product type	Tonnes	% of total textiles found in residual stream	% of all waste disposed in residual stream
Clothing	259,500	47%	2%
Other non-clothing textiles	165,640	30%	1.3%
Shoes, bags & belts	121,250	22%	0.8%
Total textiles	**546,390**	**-**	**4.2%**

Table 12: Tonnes (T) of textiles disposed of in household residual waste bins in UK in 2021.

Kerbside residual waste treatment

In England in 2020 to 2021, once an item was disposed of in the household residual waste bin and collected at kerbside by local authorities, the majority of waste (84%) was sent to incineration with energy recovery[55] and 11% headed to landfill. A landfill tax must be paid by local authorities and waste management companies to place volumes in landfill. This has an estimated value of £10.2 million, or per kilo £0.1.

* Other includes incineration without energy recovery at 0.49%. Other methods are not listed in WDF.



Incineration with energy recovery **84%**



Landfill **11%**



Other* **5%**

54 WRAP, Unpublished, England Waste Synthesis. English data scaled up to estimate UK tonnages
55 Waste Data Flow Q100. 2020-2021. England only

HWRC residual waste

164.9 kilotonnes of textiles were disposed of in residual waste banks at HWRCs in UK in 2021[56] . This calculates to 10% of all waste discarded in the HWRCs residual stream.

'Other non-clothing textiles', which primarily includes household textiles, accounts for the largest proportion of textiles by product type at HWRCs at 49%, followed by clothing at 37%, - see Table 13 for breakdown.

Product type	Tonnes	% of textiles in HWRC residual stream
Other non-clothing textiles	80,600	49%
Clothing	61,760	37%
Shoes, bags & belts	22,500	14%
Total textiles	**164,900**	**–**

Table 13: Tonnes (T) of textiles disposed of in HWRC residual waste banks in UK (scaled) in 2021.



56 WRAP, Unpublished, England Waste Synthesis. English data scaled up to estimate UK tonnages.



LOOKING AHEAD

Exploring the dynamics of research and development investment and the emergence of innovative materials within the UK textile and apparel sector.

New, innovative materials, processes, and technology

The Organisation for Economic Co-operation and Development (OECD) classes R&D spending into three different categories:

- "Basic research, which is experimental or theoretical work undertaken primarily to acquire new knowledge of the underlying foundations of phenomena and observable facts, without any particular application or use in view.
- Applied research, which is original investigation undertaken in order to acquire new knowledge. It is, however, directed primarily towards a specific, practical aim or objective.
- Experimental development, which is systematic work, drawing on knowledge gained from research and practical experience and producing additional knowledge, which is directed to producing new products or processes or to improving existing products or processes[57]."

Investment in Research & Development (R&D) in textiles and apparel in the UK

Compared to other creative industries R&D in the textiles and apparel industry has been limited due to late adoption of technology, lack of innovation, and structural constraints[58].

A survey revealed that 82% of businesses in the fashion, textiles, and technology ecosystem are SMEs, making it challenging for them to invest in R&D[59].

From 2006 to 2016, R&D expenditure in the UK textiles, apparel, and leather industries remained stable at an average of £17 million per year[60]. Although investment peaked at £39 million in 2017, it declined to £29 million in 2020. In comparison to the total R&D expenditure across all industries in the UK, which reached £44 billion in 2020 and increased to £46.9 billion in 2021, the textile and apparel manufacturing sector allocates a relatively small portion of funds to R&D.

The UK textile, clothing, and leather manufacturing sector prioritises applied research over basic and experimental development, resulting in limited innovation and technology adoption. Increased investment in experimental development could lead to product and process improvements[61]. Funding for R&D primarily comes from businesses' own funds, followed by government and overseas investments[62].

57 R&D in the creative industries | Nesta
58 Business of Fashion, Textiles & Technology: Mapping the UK Fashion, Textiles and Technology Ecosystem (arts.ac.uk)
59 House of Commons, Business Statistics,(2022).
60 Textiles & clothing businesses: R&D expenditure 2006-2020 | Statista
61 The UK Fashion and Textile Association (2020) Compendium of Industry Analysis and Trends. UKFT.
62 Research and development in the UK: flow of funds explained - Office for National Statistics (ons.gov.uk)

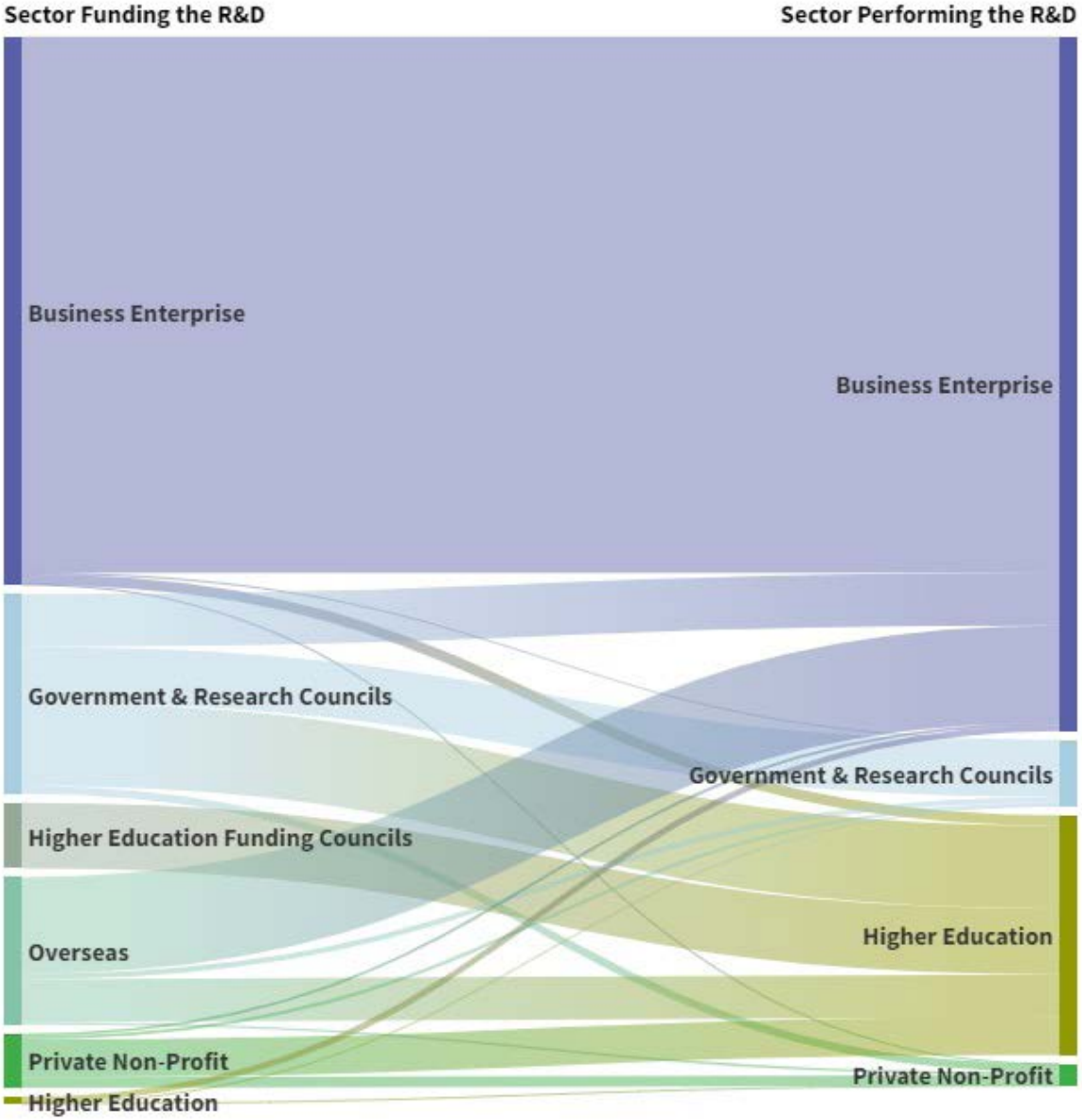


Figure 0: Flow of R&D development funding in the UK. Source: Office for National Statistics (2022).

Investment in Textile and Apparel Manufacturing

Gross Fixed Capital Formation (GFCF)[63] for apparel manufacturers remained stable at an annual average of £40 million from 2008 to 2018[64], while the textile manufacturing sector experienced more volatility, ranging from £105 million in 2010 to £245 million in 2017[65]. However, both sectors have seen significant declines in GFCF since 2018, with textile manufacturing falling by £277 million and apparel manufacturing by £96 million. In 2019, investment in textile manufacturing reached – £1 million and -£49 million for apparel manufacturing, indicating investment levels lower than depreciation. This decline in investment can be attributed to factors such as Brexit, Covid-19, and economic recession[66].

63 A measure of the net investment in capital (with deductions made for disposals) including machinery, IT systems, intellectual property, etc.

64 Apparel manufacturers refer to those producing clothing.

65 The UK Fashion and Textile Association (2020) Compendium of Industry Analysis and Trends. UKFT.

66 Business of Fashion, Textiles & Technology: Mapping the UK Fashion, Textiles and Technology Ecosystem (arts.ac.uk)

Wrap's Circular Design Toolkit[67] glossary defines next-gen materials as "Materials that are designed to replace conventional materials, and use a variety of innovative approaches to replicate the aesthetics and performance of their conventional counterparts". Next-generation materials use a variety of biomimicry approaches to replicate the aesthetics and performance of petroleum-based synthetics or animal-based counterparts[68].

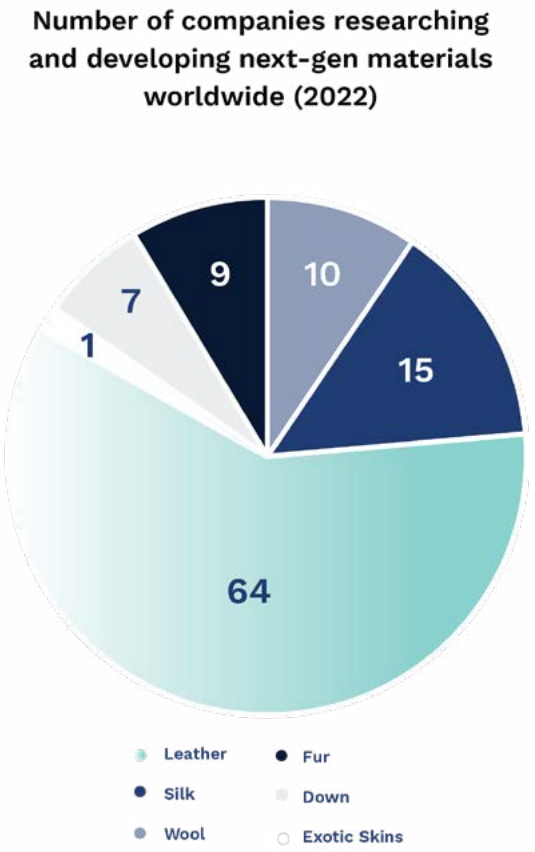


Figure 11. Number of companies researching and developing next-gen materials worldwide.

Innovative materials and processes in the UK

By main input, next-gen materials can be categorised into[69]:

- Plant-derived: applies to virgin or waste or by product plant matter.
- Mycelium: applies to root-like structure of fungal species called mycelium.
- Cultivated animal cells: applies to tissue engineering approached to grow animal cell constructs in the laboratory.
- Microbe-derived: applies to cellular engineering approached such as cell culture or fermentation processes to produce products such as proteins and biopolymers.
- Recycled materials: applies to recycled plastic or recycled textile feedstock as a main input.
- Blend: applies to blend of components not well-captured by any of the above categories.

At a global level, there are 102 companies focused exclusively in researching and developing next-gen materials, with $457 million invested in 2022[70] . 67 of these companies are trying to create alternatives to leather[71]. At a UK level, there are 8 companies that exclusively focus on the research and development of next-gen materials[72].

67 WRAP, Circular Design Toolkit, 2023.
68 MIC-23-Report-230127-1.pdf (materialinnovation.org)
69 2021 State of The Industry Report: Next-Gen Materials (2022). Material Innovation Initiative.
70 2022 State of The Industry Report: Next-Gen Materials (2023). Material Innovation Initiative.
71 Next-gen material companies globally by type 2021 | Statista
72 2022 State of The Industry Report: Next-Gen Materials (2023). Material Innovation Initiative.

Supply chain mapping and transparency

Consumer values are shifting towards socially and environmentally conscious purchasing due to the pandemic and climate emergency, with 26% of UK consumers prioritising the environment when making purchases[73]. In 2020, the UK fashion sector invested £623.9 thousand in improving supply chain ethics and sustainability, focusing on packaging, logistics, industry memberships, and supplier auditing[74]. However, progress on transparency in the global fashion industry is slow, with brands disclosing policies and commitments more than outcomes and impacts. UK-based brands scored higher in governance and transparency compared to the global average[75]. Calls for supply chain mapping and public disclosure are increasing, but only 58 brands in the UK have disclosed through the Open Apparel Registry[76]. Legislation that could mandate greater public disclosure is expected in the near future[77].

	UK Average	Global Average
Governance	48.2%	34%
Transparency	30.6%	38%

Table 14: Comparison of supply chain commitment scores by FTI, UK and Global Average.
Source: FTI (2022).



73 The new normal is about shopping better not more | EY UK
74 Reshaping Retail (barclayscorporate.com)
75 Fashion Transparency Index 2022 (2022). Fashion Revolution.
76 Open Supply Hub
77 Sustainability Reporting: UK - Makersite GmbH



WRAP is a climate action NGO working around the globe to tackle the causes of the climate crisis and give the planet a sustainable future. Our vision is a thriving world in which climate change is no longer a problem. We believe that our natural resources should not be wasted and that everything we use should be re-used and recycled. We bring together and work with governments, businesses and individuals to ensure that the world's natural resources are used more sustainably. Our core purpose is to help tackle climate change and protect our planet by changing the way things are produced, consumed and disposed of.

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